

Generalized Additive Model for Council District III (2023 - 2025 Sales)

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Contents

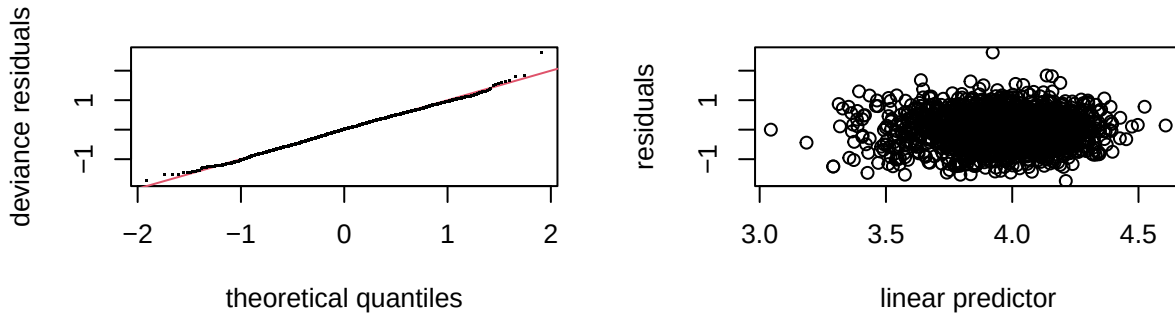
Model I	1
Model II	3
Model III	5
Model IV	8

Model I

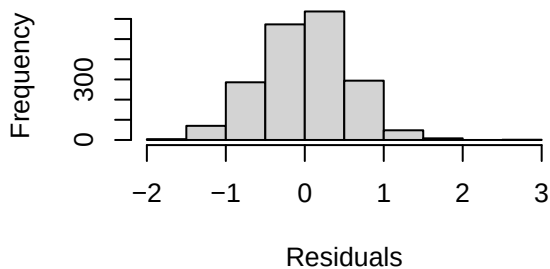
```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## ln_sppsf ~ arterial + single_family1story + half_duplex1story +
##   half_duplex2story + smallest + small + large + largest +
##   two_plus_baths + powder_room + good + fair + poor + very_poor_unsound +
##   qabove_avg + qbelow_avg + s(sresb_yearbuilt, bs = "cs", k = 10) +
##   crawl + slab + s(ln_garagespaces, bs = "cs", k = 4) + fireplace +
##   hot_water + elec_baseboard + wall + central_air + s(months,
##   bs = "cs", k = 10)
##
## Parametric coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    3.943533   0.050735  77.727 < 2e-16 ***
## arterial        0.106430   0.049258   2.161 0.030847 *
## single_family1story 0.099291   0.045561   2.179 0.029431 *
## half_duplex1story  0.167504   0.252728   0.663 0.507549
## half_duplex2story -0.896527   0.554073  -1.618 0.105815
## smallest        0.151365   0.043666   3.466 0.000539 ***
## small          0.029059   0.032060   0.906 0.364845
## large          -0.180894   0.045030  -4.017 6.12e-05 ***
## largest        -0.074324   0.117546  -0.632 0.527266
## two_plus_baths  -0.031980   0.159715  -0.200 0.841322
## powder_room     -0.058363   0.083060  -0.703 0.482350
## good            0.080207   0.117452   0.683 0.494759
## fair           -0.209018   0.028529  -7.327 3.48e-13 ***
## poor           -0.440450   0.059064  -7.457 1.34e-13 ***
```

```
## very_poor_unsound      0.009463    0.185789    0.051 0.959385
## qabove_avg             0.000000    0.000000      NaN      NaN
## qbelow_avg            -0.048169    0.029327   -1.642 0.100661
## crawl                 0.105228    0.069200    1.521 0.128517
## slab                  0.269931    0.094856    2.846 0.004479 **
## fireplace             0.076836    0.034711    2.214 0.026975 *
## hot_water             0.172151    0.061434    2.802 0.005127 **
## elec_baseboard        0.011164    0.556640    0.020 0.984000
## wall                  -0.233238    0.202138   -1.154 0.248704
## central_air           0.119697    0.036080    3.318 0.000925 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##              edf Ref.df    F  p-value
## s(sresb_yearbuilt) 0.8766     9 0.320   0.0524 .
## s(ln_garagespaces) 1.4281     3 5.871 3.25e-06 ***
## s(months)          1.6081     9 2.024 1.23e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Rank: 44/45
## R-sq.(adj) =  0.111   Deviance explained = 12.3%
## -REML = 1611.1   Scale est. = 0.30324   n = 1921
```

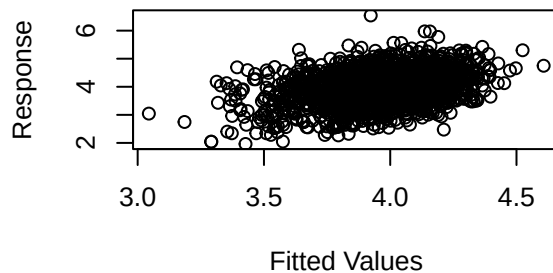
Resids vs. linear pred.



Histogram of residuals



Response vs. Fitted Values



```
##
## Method: REML   Optimizer: outer newton
## full convergence after 8 iterations.
## Gradient range [-1.375985e-06,2.090895e-08]
## (score 1611.137 & scale 0.3032418).
```

```
## Hessian positive definite, eigenvalue range [0.3113719,948.5014].
## Model rank = 44 / 45
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##          k'    edf k-index p-value
## s(sresb_yearbuilt) 9.000 0.877    0.88 <2e-16 ***
## s(ln_garagespaces) 3.000 1.428    0.84 <2e-16 ***
## s(months)          9.000 1.608    0.90 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## # A tibble: 1 x 7
##       n median_ratio mean_ratio wmean_ratio prd  cod  prb
##   <int>      <dbl>      <dbl>      <dbl> <dbl> <dbl> <dbl>
## 1  1921      0.000884      0.00106      0.000766  1.38  53.1 -1.12
```

Table 1: District 3 LN_SPPSF NLM I Assessing Coefficients

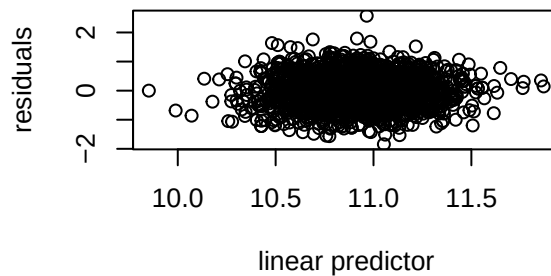
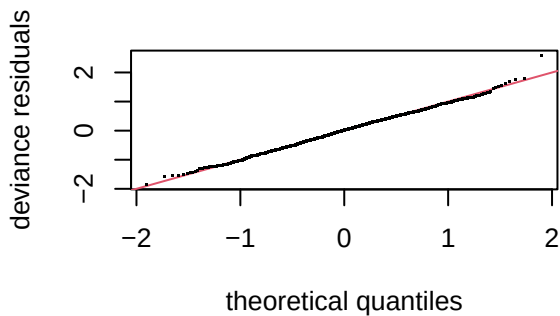
n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
1921	0.0009574	0.0011393	0.0008231	1.384192	52.59095	-0.0010544

Model II

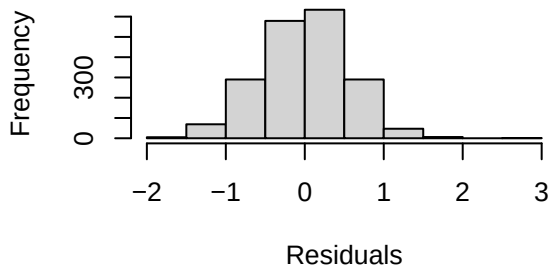
```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## ln_price ~ s(ln_landratio, bs = "cs", k = 10) + s(ln_base_bldgsize_ratio,
##   bs = "cs", k = 10) + arterial + single_family1story + half_duplex1story +
##   half_duplex2story + smallest + small + large + largest +
##   two_plus_baths + powder_room + good + fair + poor + very_poor_unsound +
##   qabove_avg + qbelow_avg + s(sresb_yearbuilt, bs = "cs", k = 10) +
##   crawl + slab + s(ln_garagespaces, bs = "cs", k = 4) + fireplace +
##   hot_water + elec_baseboard + wall + central_air + s(months,
##   bs = "cs", k = 10)
##
## Parametric coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  10.947562   0.054674  200.232 < 2e-16 ***
## arterial      0.107970   0.049036   2.202  0.02780 *
## single_family1story 0.075845   0.045875   1.653  0.09843 .
## half_duplex1story  0.061692   0.254925   0.242  0.80880
## half_duplex2story -1.094493   0.553113  -1.979  0.04799 *
## smallest      -0.052345   0.093100  -0.562  0.57402
## small        -0.077387   0.047717  -1.622  0.10501
## large        -0.059943   0.062061  -0.966  0.33423
## largest       0.135472   0.142730   0.949  0.34267
## two_plus_baths  0.091338   0.173965   0.525  0.59962
```

```
## powder_room      0.011749  0.088363  0.133  0.89424
## good             0.092854  0.116923  0.794  0.42721
## fair            -0.213542  0.028467 -7.501 9.67e-14 ***
## poor            -0.445865  0.058898 -7.570 5.80e-14 ***
## very_poor_unsound -0.004798  0.184957 -0.026 0.97931
## qabove_avg       0.000000  0.000000   NaN   NaN
## qbelow_avg       -0.053154  0.029430 -1.806 0.07106 .
## crawl            0.090569  0.071832  1.261 0.20752
## slab             0.247339  0.095821  2.581 0.00992 **
## fireplace        0.092979  0.035585  2.613 0.00905 **
## hot_water         0.174821  0.061964  2.821 0.00483 **
## elec_baseboard   -0.002060  0.554167 -0.004 0.99704
## wall             -0.245864  0.201304 -1.221 0.22210
## central_air       0.116638  0.036061  3.234 0.00124 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##              edf Ref.df    F  p-value
## s(ln_landratio)  3.8535     9 1.770  0.0012 **
## s(ln_base_bldgsize_ratio) 1.4428     9 0.655  0.0142 *
## s(sresb_yearbuilt)  0.8787     9 0.347  0.0440 *
## s(ln_garagespaces)  1.3910     3 6.328 9.33e-06 ***
## s(months)          1.6887     9 1.961 6.48e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Rank: 62/63
## R-sq.(adj) =  0.165   Deviance explained = 17.9%
## -REML =    1608   Scale est. = 0.29984    n = 1921
```

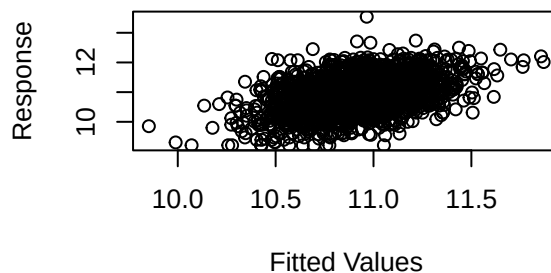
Resids vs. linear pred.



Histogram of residuals



Response vs. Fitted Values



```
##
## Method: REML    Optimizer: outer newton
## full convergence after 8 iterations.
## Gradient range [-6.042015e-07,3.665874e-07]
## (score 1608.028 & scale 0.2998402).
## Hessian positive definite, eigenvalue range [0.2957174,948.5059].
## Model rank = 62 / 63
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##              k'   edf k-index p-value
## s(ln_landratio)    9.000 3.853    0.88 <2e-16 ***
## s(ln_base_bldgsize_ratio) 9.000 1.443    0.90 <2e-16 ***
## s(sresb_yearbuilt)    9.000 0.879    0.88 <2e-16 ***
## s(ln_garagespaces)    3.000 1.391    0.85 <2e-16 ***
## s(months)            9.000 1.689    0.90 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## # A tibble: 1 x 7
##       n median_ratio mean_ratio wmean_ratio prd  cod  prb
##   <int>      <dbl>      <dbl>      <dbl> <dbl> <dbl> <dbl>
## 1  1921          0.911          1.08          0.805  1.34  49.5 -1.25
```

Table 2: District 3 LN_PRICE (No ECFs) NLM I Assessing Coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
1921	0.9853436	1.159144	0.8657068	1.338957	48.98082	-0.9723093

Model III

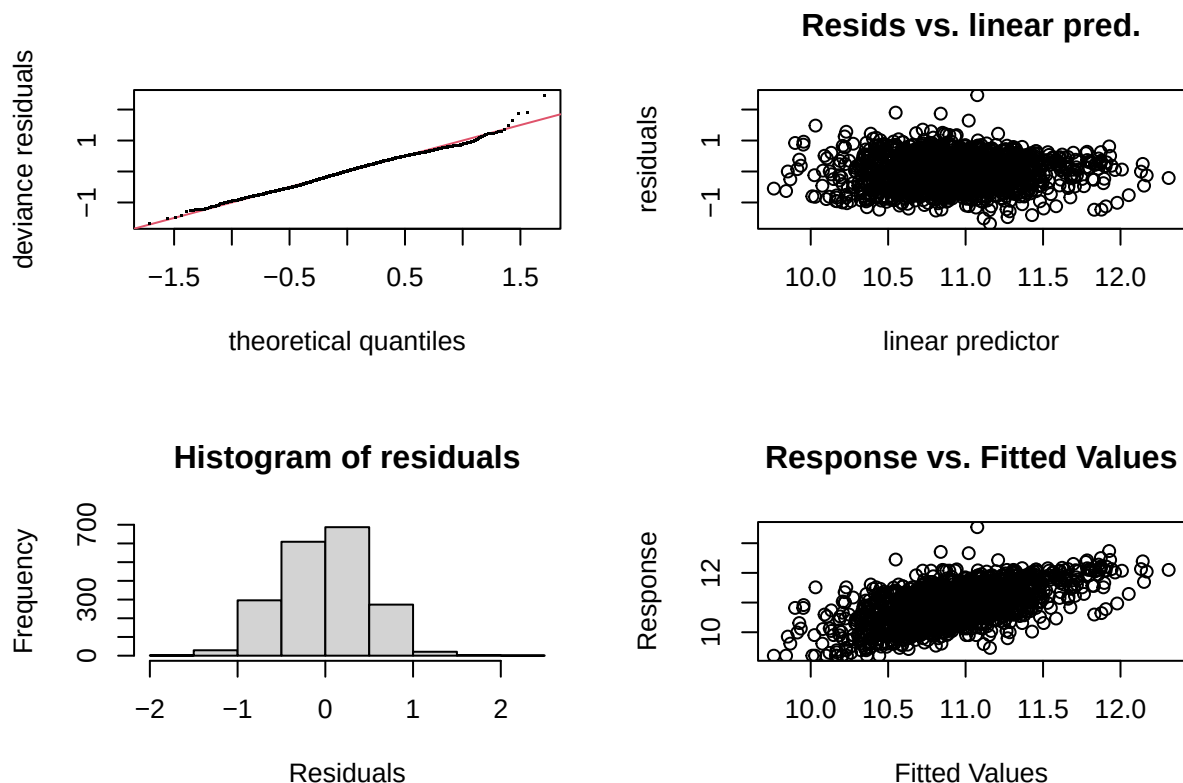
```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## ln_price ~ s(ln_landratio, bs = "cs", k = 10) + s(ln_base_bldgsize_ratio,
##   bs = "cs", k = 10) + arterial + single_family1story + half_duplex1story +
##   half_duplex2story + smallest + small + large + largest +
##   two_plus_baths + powder_room + good + fair + poor + very_poor_unsound +
##   qabove_avg + qbelow_avg + s(sresb_yearbuilt, bs = "cs", k = 10) +
##   crawl + slab + s(ln_garagespaces, bs = "cs", k = 4) + fireplace +
##   hot_water + elec_baseboard + wall + central_air + s(months,
##   bs = "cs", k = 10) + ecf
##
## Parametric coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    10.742408   0.106407 100.956 < 2e-16 ***
## arterial         0.037285   0.045121   0.826 0.408716
## single_family1story 0.070707   0.042273   1.673 0.094571 .
```

```

## half_duplex1story -0.064087 0.231464 -0.277 0.781905
## half_duplex2story -0.701045 0.522049 -1.343 0.179477
## smallest -0.014729 0.081592 -0.181 0.856766
## small -0.045530 0.042485 -1.072 0.283998
## large 0.020162 0.054896 0.367 0.713450
## largest 0.190670 0.126250 1.510 0.131147
## two_plus_baths 0.127023 0.155132 0.819 0.412999
## powder_room 0.002626 0.079323 0.033 0.973595
## good 0.067517 0.106749 0.632 0.527149
## fair -0.169405 0.026370 -6.424 1.68e-10 ***
## poor -0.383007 0.053914 -7.104 1.72e-12 ***
## very_poor_unsound 0.236107 0.168566 1.401 0.161477
## qabove_avg 0.000000 0.000000 NaN NaN
## qbelow_avg -0.095248 0.030440 -3.129 0.001781 **
## crawl -0.072875 0.066307 -1.099 0.271892
## slab 0.038013 0.088394 0.430 0.667218
## fireplace 0.040341 0.033597 1.201 0.230019
## hot_water 0.082707 0.057984 1.426 0.153926
## elec_baseboard -0.052118 0.501597 -0.104 0.917257
## wall -0.453058 0.182660 -2.480 0.013214 *
## central_air 0.093339 0.033108 2.819 0.004865 **
## ecf3R302 0.585492 0.195223 2.999 0.002744 **
## ecf3R303 0.143349 0.101113 1.418 0.156442
## ecf3R304 -0.203382 0.117857 -1.726 0.084572 .
## ecf3R305 -0.129762 0.190909 -0.680 0.496773
## ecf3R306 0.735611 0.117110 6.281 4.17e-10 ***
## ecf3R307 0.289999 0.101147 2.867 0.004189 **
## ecf3R308 -0.079342 0.127169 -0.624 0.532763
## ecf3R309 -0.071365 0.131553 -0.542 0.587554
## ecf3R310 0.978157 0.109466 8.936 < 2e-16 ***
## ecf3R311 0.256698 0.130247 1.971 0.048888 *
## ecf3R312 0.013916 0.106138 0.131 0.895703
## ecf3R313 0.043908 0.109111 0.402 0.687424
## ecf3R314 0.404087 0.114300 3.535 0.000417 ***
## ecf3R315 0.362712 0.108234 3.351 0.000821 ***
## ecf3R316 0.001285 0.114916 0.011 0.991078
## ecf3R317 -0.220996 0.183592 -1.204 0.228844
## ecf3R318 0.373261 0.198689 1.879 0.060453 .
## ecf3R319 0.129962 0.099354 1.308 0.191009
## ecf3R320 0.268774 0.103275 2.602 0.009328 **
## ecf3R321 -0.111396 0.117585 -0.947 0.343578
## ecf3R322 0.474192 0.109656 4.324 1.61e-05 ***
## ecf3R323 0.319824 0.103449 3.092 0.002020 **
## ecf3R324 -0.081506 0.127714 -0.638 0.523425
## ecf3R325 0.239077 0.111366 2.147 0.031940 *
## ecf3R326 -0.472363 0.134054 -3.524 0.000436 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##          edf Ref.df      F p-value
## s(ln_landratio) 0.888      9 0.343 0.03941 *
## s(ln_base_bldgsize_ratio) 1.195      9 0.503 0.02407 *
## s(sresb_yearbuilt) 1.726      9 1.033 0.00463 **
## s(ln_garagespaces) 1.306      3 3.093 0.00102 **
## s(months) 1.897      9 3.597 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##

```

```
## Rank: 87/88
## R-sq.(adj) = 0.324   Deviance explained = 34.3%
## -REML = 1433.4   Scale est. = 0.24289   n = 1921
```



```
##
## Method: REML   Optimizer: outer newton
## full convergence after 8 iterations.
## Gradient range [-1.425338e-05,3.129265e-06]
## (score 1433.359 & scale 0.2428942).
## Hessian positive definite, eigenvalue range [0.2943905,936.0028].
## Model rank = 87 / 88
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##           k'   edf k-index p-value
## s(ln_landratio)      9.000 0.888   0.98  0.225
## s(ln_base_bldgsize_ratio) 9.000 1.195   0.94  0.005 **
## s(sresb_yearbuilt)     9.000 1.726   0.97  0.090 .
## s(ln_garagespaces)     3.000 1.306   0.98  0.195
## s(months)             9.000 1.897   0.98  0.175
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

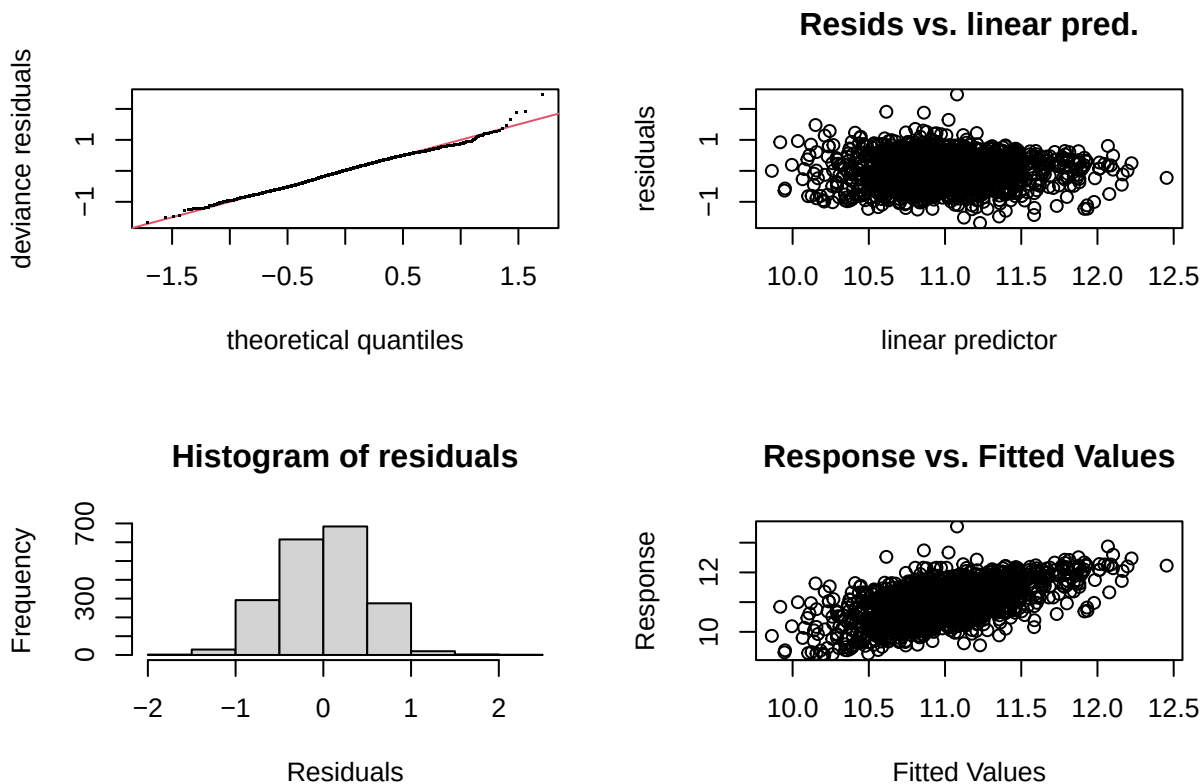
## # A tibble: 1 x 7
##       n median_ratio mean_ratio wmean_ratio prd   cod   prb
##   <int>      <dbl>      <dbl>      <dbl> <dbl> <dbl> <dbl>
## 1  1921        0.916        1.05        0.832  1.26  43.6 -0.752
```

Model IV

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## ln_tasp ~ s(ln_landratio, bs = "cs", k = 10) + s(ln_base_bldgsize_ratio,
##       bs = "cs", k = 10) + arterial + single_family1story + half_duplex1story +
##       half_duplex2story + smallest + small + large + largest +
##       two_plus_baths + powder_room + good + fair + poor + very_poor_unsound +
##       qabove_avg + qbelow_avg + s(sresb_yearbuilt, bs = "cs", k = 10) +
##       crawl + slab + s(ln_garagespaces, bs = "cs", k = 4) + fireplace +
##       hot_water + elec_baseboard + wall + central_air + ecf + s(months,
##       bs = "cs", k = 10)
##
## Parametric coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  10.808847   0.106204 101.774 < 2e-16 ***
## arterial      0.037879   0.045109   0.840 0.401168
## single_family1story 0.079438   0.041305   1.923 0.054607 .
## half_duplex1story -0.050705   0.230992  -0.220 0.826276
## half_duplex2story -0.690967   0.521787  -1.324 0.185588
## smallest      -0.012073   0.082037  -0.147 0.883015
## small         -0.046048   0.042653  -1.080 0.280456
## large         0.020583   0.055159   0.373 0.709081
## largest       0.192107   0.126706   1.516 0.129646
## two_plus_baths 0.124967   0.155493   0.804 0.421684
## powder_room   0.001854   0.079433   0.023 0.981378
## good          0.070166   0.106711   0.658 0.510920
## fair         -0.168846   0.026360  -6.405 1.89e-10 ***
## poor         -0.381245   0.053877  -7.076 2.09e-12 ***
## very_poor_unsound 0.243561   0.168265   1.447 0.147931
## qabove_avg     0.000000   0.000000      NaN      NaN
## qbelow_avg    -0.094895   0.030434  -3.118 0.001848 **
## crawl         -0.074107   0.066279  -1.118 0.263662
## slab          0.038821   0.088373   0.439 0.660502
## fireplace     0.039947   0.033571   1.190 0.234217
## hot_water     0.082212   0.057965   1.418 0.156270
## elec_baseboard -0.071237   0.500856  -0.142 0.886913
## wall          -0.450792   0.182637  -2.468 0.013667 *
## central_air    0.092015   0.033079   2.782 0.005462 **
## ecf3R302       0.587125   0.195086   3.010 0.002651 **
## ecf3R303       0.143717   0.101066   1.422 0.155189
## ecf3R304      -0.205188   0.117815  -1.742 0.081740 .
## ecf3R305      -0.127406   0.190892  -0.667 0.504584
## ecf3R306       0.735902   0.117081   6.285 4.06e-10 ***
## ecf3R307       0.290023   0.101062   2.870 0.004154 **
## ecf3R308      -0.080285   0.127131  -0.632 0.527782
## ecf3R309      -0.074027   0.131425  -0.563 0.573321
## ecf3R310       0.974572   0.109343   8.913 < 2e-16 ***
## ecf3R311       0.254641   0.130172   1.956 0.050591 .
## ecf3R312       0.013018   0.106043   0.123 0.902310
## ecf3R313       0.044014   0.109013   0.404 0.686443
## ecf3R314       0.404061   0.114231   3.537 0.000414 ***
## ecf3R315       0.361585   0.108193   3.342 0.000848 ***
## ecf3R316       0.002283   0.114876   0.020 0.984146
## ecf3R317      -0.217985   0.183530  -1.188 0.235087
## ecf3R318       0.371176   0.198606   1.869 0.061792 .
```



```
## ecf3R319          0.130930  0.099225  1.320 0.187158
## ecf3R320          0.269291  0.103164  2.610 0.009118 **
## ecf3R321         -0.109263  0.117461  -0.930 0.352383
## ecf3R322          0.474793  0.109590  4.332 1.55e-05 ***
## ecf3R323          0.321158  0.103318  3.108 0.001909 **
## ecf3R324         -0.081359  0.127630  -0.637 0.523905
## ecf3R325          0.235629  0.111267  2.118 0.034333 *
## ecf3R326         -0.472147  0.133853  -3.527 0.000430 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##              edf Ref.df      F p-value
## s(ln_landratio)    0.87544     9 0.330 0.04231 *
## s(ln_base_bldgsize_ratio) 1.22855     9 0.535 0.02095 *
## s(sresb_yearbuilt)    1.71829     9 1.015 0.00499 **
## s(ln_garagespaces)    1.30443     3 3.067 0.00107 **
## s(months)            0.06276     9 0.007 0.31053
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Rank: 87/88
## R-sq.(adj) =  0.315   Deviance explained = 33.4%
## -REML = 1431.1   Scale est. = 0.2429    n = 1921
```



```
##
## Method: REML   Optimizer: outer newton
## full convergence after 10 iterations.
## Gradient range [-0.0007441749,0.002862354]
## (score 1431.086 & scale 0.2428988).
```

```
## Hessian positive definite, eigenvalue range [0.002353907,935.999].
## Model rank = 87 / 88
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##           k'      edf k-index p-value
## s(ln_landratio) 9.0000 0.8754   0.98  0.210
## s(ln_base_bldgsize_ratio) 9.0000 1.2285   0.94  0.005 **
## s(sresb_yearbuilt) 9.0000 1.7183   0.97  0.140
## s(ln_garagespaces) 3.0000 1.3044   0.98  0.285
## s(months) 9.0000 0.0628   0.98  0.185
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## # A tibble: 1 x 7
##       n median_ratio mean_ratio wmean_ratio prd  cod  prb
##   <int>      <dbl>      <dbl>      <dbl> <dbl> <dbl> <dbl>
## 1  1921      0.985      1.13      0.894  1.26  43.3 -0.766
```